

Original Article

Arbuscular mycorrhizal fungi on the development and copper content in corn and sorghum plants

Fungos micorrízicos arbusculares no desenvolvimento e teor de cobre em plantas de milho e sorgo

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Abstract

The concentration of copper in the soil increased with the intensification of agricultural activities, mainly in grape production areas and orchards as a result of the application of pesticides. Arbuscular mycorrhizal fungi make up the microbial biomass of the soil and appear as an alternative to be researched for the development of plants in an environment contaminated with copper. The purpose of this pot study was to analyze the influence of arbuscular mycorrhizal fungi on the development and content of copper in corn and sorghum plants. Soil treatments were: without inoculum (control) and two arbuscular mycorrhizal (*Acaulospora scrobiculata* and *Rhizoglossum clarum*) and five doses of copper (0, 100, 200, 300, and 400 mg Cu kg⁻¹ soil); with seven repetitions. Plant height, stem diameter, number of tillers, root volume, shoot and root dry weight yields, shoot, root and grain Cu concentrations, pseudo-total soil Cu, percentage of mycorrhizal colonization and relative mycorrhizal efficiency index in reducing Cu concentration in root and shoot of corn and sorghum were evaluated. Morphological parameters of sorghum and corn were reduced with at high Cu doses in the soil, and the inoculation with *Acaulospora scrobiculata* and *Rhizoglossum clarum* resulted in greater development and lower Cu concentration in the dry mass of the shoot and root parts sorghum and corn plants.

Keywords: *Sorghum bicolor* L., phytostabilization, heavy metal, mycorrhiza, polluted soil, *Zea mays* L.

Resumo

A concentração de cobre no solo aumentou com a intensificação das atividades agrícolas, principalmente em áreas de produção de uva e pomares, como resultado da aplicação de pesticidas. Fungos micorrízicos arbusculares compõem a biomassa microbiana do solo e aparecem como uma alternativa a ser pesquisada para o desenvolvimento de plantas em ambiente contaminado com cobre. O objetivo deste estudo em vasos foi analisar a influência dos fungos micorrízicos arbusculares no desenvolvimento e conteúdo de cobre em plantas de milho e sorgo. Os tratamentos do solo foram: sem inóculo (controle) e dois micorrízicos arbusculares (*Acaulospora scrobiculata* e *Rhizoglossum clarum*) e cinco doses de cobre (0, 100, 200, 300 e 400 mg Cu kg⁻¹ solo); com sete repetições. Foram avaliados altura de planta, diâmetro do caule, número de perfilhos, volume de raiz, produtividade de matéria seca de parte aérea e raiz, concentrações de Cu em parte aérea, raiz e grão, Cu pseudo-total do solo, porcentagem de colonização micorrízica e índice de eficiência micorrízica relativa na redução da concentração de Cu em raiz e parte aérea de milho e sorgo. Os parâmetros morfológicos de sorgo e milho foram reduzidos com altas doses de Cu no solo, e a inoculação com *Acaulospora scrobiculata* e *Rhizoglossum clarum* resultou em maior desenvolvimento e menor concentração de Cu na massa seca das partes aérea e radicular de plantas de sorgo e milho.

Palavras-chave: *Sorghum bicolor* L., fitoestabilização, metal pesado, micorriza, solo poluído, *Zea mays* L.

1. Introduction

Copper (Cu) makes up the structure of minerals and is naturally present in the soil (Santana, 2020). However, as a result of the application of pesticides, their levels in the soil have increased with the duration of agricultural activity (Neaman et al., 2024), mainly in vineyard and orchard soils

(Brunetto et al., 2014; Schoffer et al., 2020), which can cause physiological, biochemical, and morpho-anatomical disorders in plants, with inactivation of cytoplasmic enzymes, oxidative stress, reduction of photosynthesis processes, compromising the growth and development of plants (Marques et al., 2018).

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However, some plant species have the ability to extract or metabolize elements that are at levels considered toxic in the soil, avoiding translocation to their aerial part (Moogouei et al., 2011) and may be an alternative for use in soils contaminated with copper. In this sense, the literature reports the possibility of using both forest species (De Marco et al., 2017; Bomfim, 2020), cover crops (Vendrusculo 2013), as well as agricultural crops such as corn (Jarausch-Wehrheim et al., 1996) and sorghum (Angelova et al., 2011). However, the use of corn and sorghum is conditioned by the concentration of copper in the grain or in the aerial biomass, which must not exceed the limits established by legislation, which are 10 mg kg⁻¹ for the grain (Brasil, 1998) and 30 mg kg⁻¹ for aerial biomass for animal feed purposes in Brazil (Brasil, 1965).

Corn (*Zea mays* L.) is a culture of great importance for the various products generated from its raw material, such as fuels, beverages, and polymers, in addition to being used in human and animal food (Contini et al., 2019). It is very demanding in nutrients, including copper, but in high concentrations it becomes phytotoxic, causing a reduction in height, leaf area, yield, and root dry mass (Barbosa et al., 2013). Sorghum (*Sorghum bicolor* L.) is considered the fifth most important cereal in the world, with cultivation intended primarily for human and animal consumption and ethanol production (Guedes et al., 2019) and in Brazil and other western countries it is used for animal feed (Taleon et al., 2012). It stands out for its rusticity and the possibility of cultivation outside the agroclimatic zoning of corn and for having greater tolerance to water deficit (Duarte 2010). It is sensitive to copper when the available levels are in high concentration in the soil, which can cause negative effects, such as reduction of stem diameter, plant height, and number of tillers (Silva 2019).

Regarding microorganisms, Arbuscular Mycorrhizal Fungi (AMF) are important components of soil microbial biomass and are involved in several processes of the soil-plant system (Rodrigues et al., 2018). These organisms have structures known as hyphae which, when associated with plants, function as an extension of the root system, allowing greater absorption of water and nutrients, providing a greater increase in their biomass and quality, in addition to promoting tolerance to abiotic stresses and chemical elements at levels considered toxic in the soil (Valadares et al., 2016; Souza et al., 2023). They are keyplayers in protecting plants against high concentrations of Cu in the soil, and can act through the accumulation of metal in non-vital organs, retention in fungal mycelium and cell walls (Ruscitti et al., 2017). Although AMF have the potential to be used in contaminated soil, this work questions whether

their association with corn and sorghum plants will result in development and lower copper levels in the aerial part of the plants when grown in soil contaminated with copper. The objective of this study was to analyze the influence of arbuscular mycorrhizal fungi on the development and copper content in corn and sorghum plants.

2. Materials and Methods

The study was carried out in a greenhouse belonging to the Department of Forestry of the Universidade Federal de Santa Maria, campus Frederico Westphalen, RS, with two experiments being simultaneously performed, for sorghum and corn. The soil used to develop the experiment was a soil constructed (substrate) from a 0-20 cm sample of soil characterized as Red Latosol (Santos 2018) cultivated under native fields. After collection, the soil was sieved in a 2 mm mesh and, to facilitate the root analysis, it was mixed with medium sand in a 50% proportion (v/v), aiming at an approximate 46% texture. A sample was taken and sent for analysis of physical and chemical attributes (Table 1) and then subjected to sterilization in an autoclave at 121 °C in 3 cycles of 30 min.

After interpretation of the analysis, the pH was adjusted to 6.5 with the use of dolomitic limestone and the fertilization was carried out with the use of mineral fertilizer, formula NPK 10-20-10, according to the recommendation of the Liming and Fertilization Manual for the States of RS and SC (CQFS, 2016), for sorghum and corn cultures. Soil contamination with Cu doses occurred 30 days before sowing, with the use of Cu sulfate (CuSO₄·5H₂O), which was homogenized by stirring in a plastic bag along with the soil.

The isolates of arbuscular mycorrhizal fungi (*Acaulospora scrobiculata* and *Rhizoglyphus clarum*) were obtained from Embrapa Agrobiologia located in Seropédica-RJ. The corn cultivar used was MORGAN 20A78 and the sorghum cultivar was AGROCERES 2501.

The seeds were previously disinfected with 2% sodium hypochlorite for 15 min and then washed in running water. At the time of sowing, 30 spores of each mycorrhizal fungus isolate were inoculated in each pot, corresponding to the treatment. The pots used in the experiment had a capacity of 5 liters, which were filled with 5 kg of soil. Sowing took place in the first fortnight of October 2021, with 4 seeds being sown per pot; the thinning of the seedlings occurred 10 days after sowing, leaving only one plant per pot. The plants were irrigated daily, with water

Table 1. Clay content of the substrate (soil + sand) and chemical analysis of the original soil collected in a native field used for sorghum and corn cultivation.

Substrate	Original Soil							
Clay content	pH	MO	P	K	Cu*	Zn	Mg	Al+H
%	1:1	%	-----mg/L-----			-----Cmolc/L----		
46	5.3	1.0	4.6	27.5	4.2	1.3	1.0	4.9

*Copper available.
Source: Author (2022)

from the public supply system, using a drip irrigation system, maintaining soil moisture at 80% of field capacity.

The design was completely randomized, with a factorial arrangement (3x5), with three inoculations of mycorrhizal fungi (*Acaulospora scrobiculata*, *Rhizoglyphus clarum* and control without inoculation), five doses of copper (0, 100, 200, 300, and 400 mg Cu kg⁻¹ soil), with seven repetitions.

At the end of the crop cycle, after 120 days of cultivation, evaluations were performed: Plant height (PH), measured from the root crown to the flag leaf, using a tape measure; stem diameter (SD), with the aid of a digital caliper; number of tillers (sorghum); and root volume (RV), with the aid of a graduated cylinder. Each plant was divided into shoot (stem, leaves, cob, straw), roots, and grains (for corn), placed in paper bags and taken to an oven at 65°C until they had constant weight. Subsequently, shoot, roots and grains were ground in a Willey mill with a 10-mesh sieve for analysis of copper contents, by means of nitro-perchloric digestion (3:1) and determination in atomic absorption spectrophotometry according to Miyazawa et al. (2009). The pseudo-total Cu contents in the soil were also determined using the 3050b methodology described by USEPA (1996) and available copper through the Mellich⁻¹ extracting solution (Sobral et al., 2013).

Mycorrhizal colonization was performed using the technique of clarification and root staining with 0.05% Trypan Blue and the percentage of colonization estimated in three replicates per plant by the grid-line method (Giovannetti and Mosse, 1980).

The mycorrhizal efficiency index was used to compare the efficiency of inoculation of mycorrhizal fungi in reducing the Cu concentration in the roots and aboveground plant parts, being calculated according to Equation 1, using the treatment without inoculation as a control:

$$MEI(\%) = \frac{\text{Without mycorrhizal}}{\text{Inoculated}} \times 100 \quad (1)$$

The results were subjected to analysis of variance with the aid of the SISVAR statistical program (Ferreira 2011) and when there was significant interaction, the effects of doses in each inoculation treatment were unfolded through regression analysis and, when not significant, the qualitative factor (inoculum) was evaluated by comparing means by the Scott-Knott test at 5% error probability and the quantitative factor (copper doses) was evaluated by regression analysis.

3. Results and Discussion

The pseudo-total soil Cu increased with the doses applied to sorghum and corn, and from the dose 67 mg Cu kg⁻¹ in the soil to sorghum and 80 mg Cu kg⁻¹ to corn (Figure 1), the values were above the maximum limit allowed for agricultural areas in Brazil established by Conama (Brasil, 2009), which is 200 mg Cu kg⁻¹ and 203 mg Cu kg⁻¹ by Fepam-RS (Porto Alegre, 2014). Copper may be available to plants when present in the soil solution in free form and also complexed with dissolved organic matter. The availability of this metal is controlled by chemical

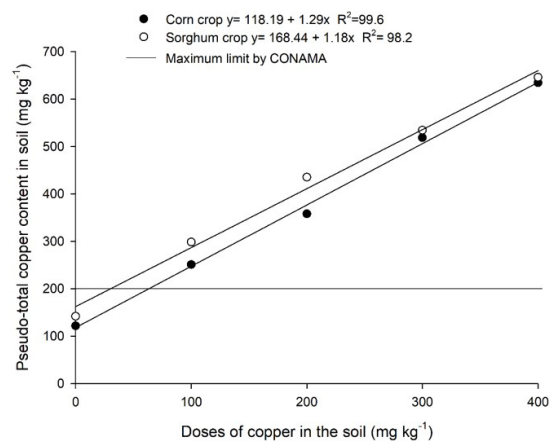


Figure 1. Pseudo-total copper content in soil cultivated with sorghum and corn.

processes such as precipitation-dissolution, adsorption-desorption and complexation (Arend, 2010).

The results showed no significant interaction between the Cu doses incorporated into the soil and inoculum sources for the following plant parameters: plant height (PH), stem diameter (SD), root volume (RV) of sorghum and corn and number of sorghum tillers. For root volume and plant height of sorghum and corn, there was a simple effect for the two sources of variation, however, stem diameter and number of tillers only differed depending on the inoculum sources (Figure 2 and Table 2).

Plant height decreased linearly with Cu doses for sorghum and corn (Figure 2A and 2B). Similar results were observed by Fruet (2018) when evaluating the forest species *Peltophorum dubium* and by Panziera et al. (2018) when analyzing black oat, in both cases, there was a reduction in plant height with Cu excess. High Cu concentrations interfere with the electron transport chain of photosystem I, reducing the production of photoassimilates and consequently decreasing apical growth (Taiz and Zeiger 2017).

Sorghum root volume decreased linearly with increasing copper doses (Figure 2C), while corn showed a quadratic response with a maximum point at the 167.50 mg Cu kg⁻¹ dose, corresponding to 93.36 cm³ of volume (Figure 2D). Silva (2019) also showed a decrease in root volume for soybean and sorghum with increasing copper doses. Cu toxicity can cause oxidative stress in plants, due to the increase in the production of highly toxic oxygen free radicals that damage cell membrane lipids and proteins (Yruela, 2009), inducing reduction in the root system. However, in some situations, when exposure to Cu becomes excessive, there may be an increase in the number of lateral roots (Bochicchio et al., 2015) and the diameter of the roots in an attempt to store Cu in the vacuoles (Juang et al., 2012).

The mycorrhizal inoculum increased plant height, stem diameter, root volume, and number of tillers in sorghum, when compared with the control, however, with no difference between the studied species, while the *R. clarum* inoculum provided greater stem diameter and

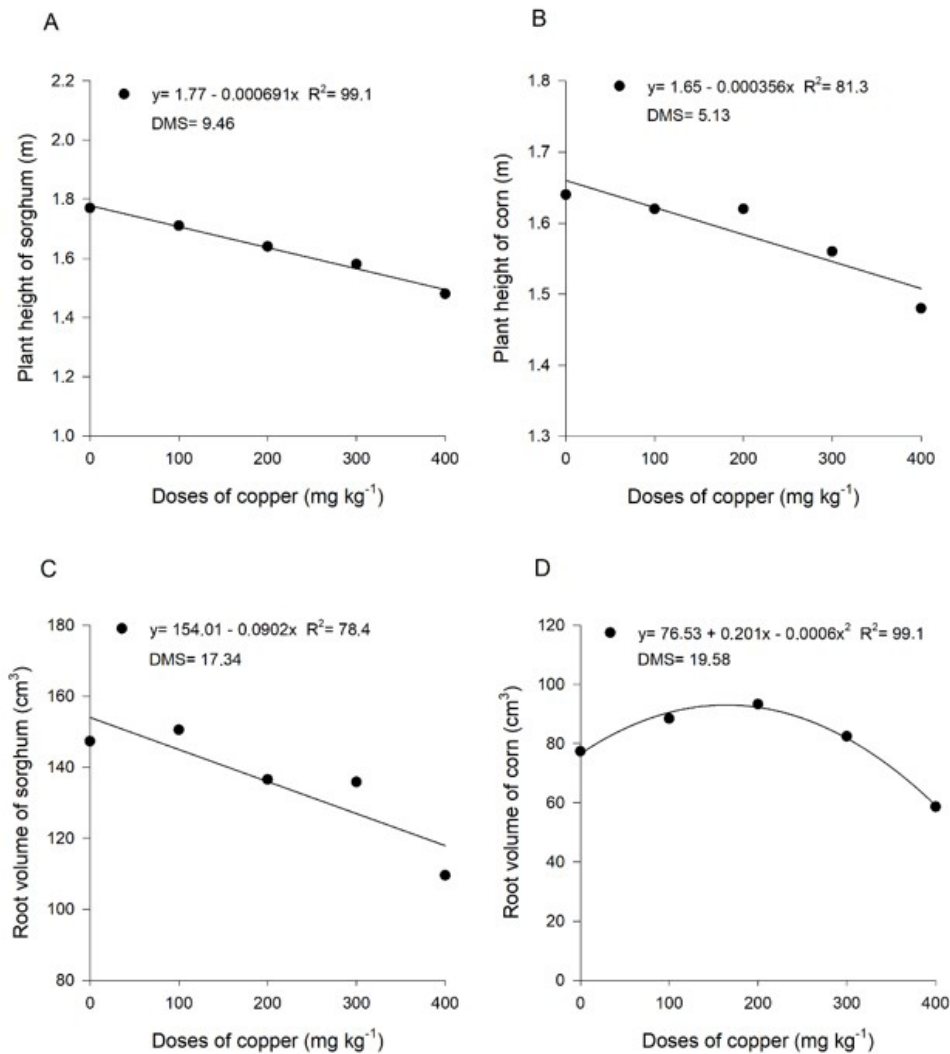


Figure 2. Plant height (PH) and root volume (RV) of sorghum and corn subjected to different doses of copper in the soil.

Table 2. Simple effect of inoculation plant height (PH), stem diameter (SD), and root volume (RV) of sorghum and corn and number of sorghum tillers.

Inoculum source	PH (m)	SD (mm)	RV (ml)	Nº tillers
	-----Sorghum-----			
<i>A. scrobiculata</i>	1.76 A*	11.33 A	142.37 A	3.57 A
<i>R. clarum</i>	1.68 A	10.95 A	142.92 A	3.58 A
Control	1.47 B	9.44 B	122.63 B	2.94 B
CV (%)	15.91	17.00	29.18	42.96
	-----Corn-----			
<i>A. scrobiculata</i>	1.65 A	14.62 B	80.45 B	nd
<i>R. clarum</i>	1.61 A	16.44 A	96.20 A	nd
Control	1.48 B	14.35 B	63.48 C	nd
CV (%)	8.69	21.19	32.95	nd

*Means followed by the same capital letter in the column do not differ from each other by the Scott-Knott test ($p < 0.05$) of error probability. nd: not determined. CV: coefficient of variation.

volume of roots in the corn (Table 2). When evaluating the effect of inoculating AMF with doses of phosphorus in corn, Gomes Junior et al. (2018) found an increase in root volume when inoculating mycorrhizal fungi. The increase in plant height and stem diameter are benefits provided by AMF and are related to the ability of these fungi to produce hyphae, which increase the volume of soil explored by the root system, favoring plant growth (Abreu et al., 2018). In addition, AMF are responsible for increasing the surface of the roots, allowing a greater capacity to absorb water and nutrients necessary for the growth and survival of plants (Cristino et al., 2019).

Sorghum shoot dry mass quadratically increased up to the estimated dose of 165 mg Cu kg⁻¹ soil (Figure 3A) and corn showed a linear reduction as a function of copper doses (Figure 3B). When the root and shoot dry mass yield decreased, can induce a higher Cu concentration in these plant parts because the dilution in the biomass is lower (Figure 3). High Cu doses reduced the dry mass of shoots, both in annual plants, such as sorghum and in tree species, such as *Erythrina crista-galli* (May et al., 2020; De Marco et al.,

2021). Excess copper can cause a reduction in photosynthetic rate, decreasing the production of photoassimilates, and reducing plant growth (Kabata-Pendias, 2011).

Dry mass of sorghum roots and corn root quadratically increased with the highest value in the doses of 192 mg Cu kg⁻¹ soil for sorghum (Figure 3C) and 171 mg Cu kg⁻¹ soil for corn (Figure 3D). This result corroborated those in the literature, which also showed a reduction in the dry mass of corn roots from a dose of 200 mg Cu kg⁻¹ (Mantovani, 2009) and a reduction in the growth of corn plants with a dose of 120 mg Cu kg⁻¹ (Tiecher et al., 2018).

There was no significant interaction for the dry mass of shoots and roots of sorghum and corn, only significant simple effects of the variation factors (Table 3). Inoculations increased shoot dry mass of sorghum and corn in relation to the control treatment but did not differ between the studied species (Table 3). The dry mass of sorghum root increased with the inoculation of the species *A. scrobiculata* with an average of 0.049 kg.

In corn, there was no difference between treatments, with an average of 0.026 kg. A positive effect of AMF

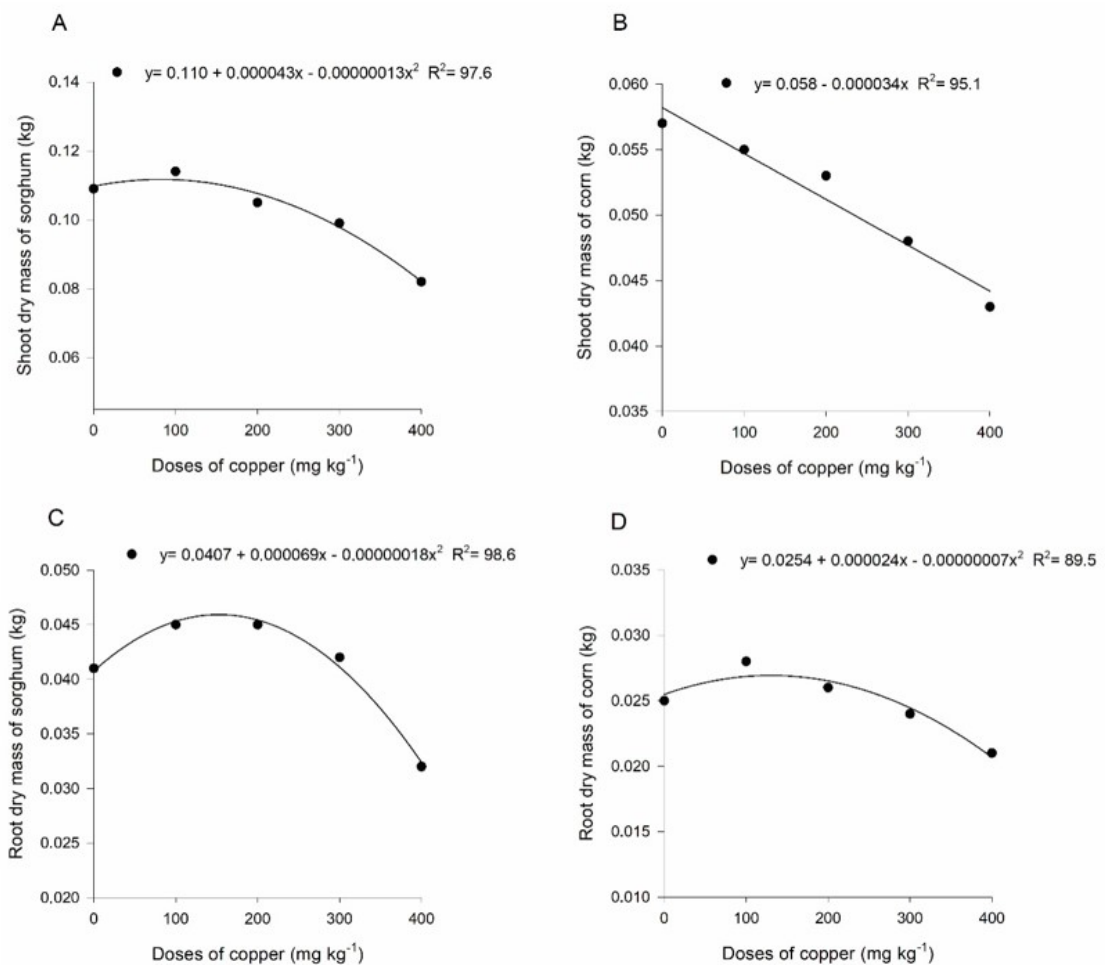


Figure 3. Shoot dry mass (SDM) of sorghum (A) and corn (B) and root dry mass (RDM) of sorghum (C) and corn (D) subjected to different doses of copper in the soil.

inoculation on shoot dry mass was also observed in forest species such as *Genipa americana* and *Schizolobium amazonicum* and in corn, cotton, beans, and soybeans (Soares et al., 2007; Brito et al., 2017; Santos et al., 2018; Salgado, 2014). In addition, Salgado (2014) showed that the species *A. scrobiculata* and *Claroideoglossum etunicatus* provided higher dry mass of corn roots. AMFs are also capable of increasing plant biomass and carbon fixation in the plants' shoot, due to greater absorption of nutrients (Braghirolli et al., 2012).

There was a significant interaction between inoculum sources and Cu doses for the percentage of mycorrhizal colonization in sorghum and corn (Figure 4). The mycorrhizal colonization of sorghum decreased linearly with increasing Cu doses in the soil, being more expressive with the inoculation of *R. clarum*, while *A. scrobiculata* showed greater root colonization at the highest Cu dose tested (Figure 4A). In corn, mycorrhizal colonization behaved quadratically for both species, and *A. scrobiculata* showed higher colonization (61.9%) with a maximum point at the

dose of 22 mg Cu kg⁻¹; however, they did not differ from each other from of the 300 mg Cu kg⁻¹ dose (Figure 4B).

When evaluating the percentage of colonization of AMF species in agricultural crops, Salgado (2014) found that *A. scrobiculata* colonized around 30 to 40% in corn and bean plants, being lower for cotton. However, Rosa (2019), working with vine seedlings, observed a 70% reduction in the mycorrhizal colonization rate with the addition of 150 mg Cu kg⁻¹ soil, thus demonstrating that high Cu doses have a toxic effect on this association.

There was no significant interaction between Cu doses and inoculum sources for the Cu concentration in the roots (CuR) and in the shoot (CuPA) of sorghum and corn and Cu concentration in the grain (CuG) of corn, presenting only a single effect for the variation factors (Figure 5 and Table 4). Root Cu concentrations for sorghum and corn increased linearly with increasing Cu doses in the soil (Figure 5). In the roots, there was also the highest Cu concentration, in relation to the other plant parts, being higher in corn with an increase of 76% as a function of the applied Cu

Table 3. Simple effect of inoculation shoot dry mass (SDM) and root dry mass (RDM) of sorghum and corn.

Inoculum source	-----Sorghum-----		-----Corn-----	
	SDM (kg plant ⁻¹)	RDM (kg plant ⁻¹)	SDM (kg plant ⁻¹)	RDM (kg plant ⁻¹)
<i>A. scrobiculata</i>	0.113 A*	0.049 A	0.056 A	0.025 A
<i>R. clarum</i>	0.105 A	0.041 B	0.057 A	0.026 A
Control	0.087 B	0.032 C	0.041 B	0.023 B
CV (%)	22.33	31.33	18.10	20.04

*Means followed by the same capital letter in the column do not differ from each other by the Scott-Knott test (p<0.05) of error probability. CV: coefficient of variation.

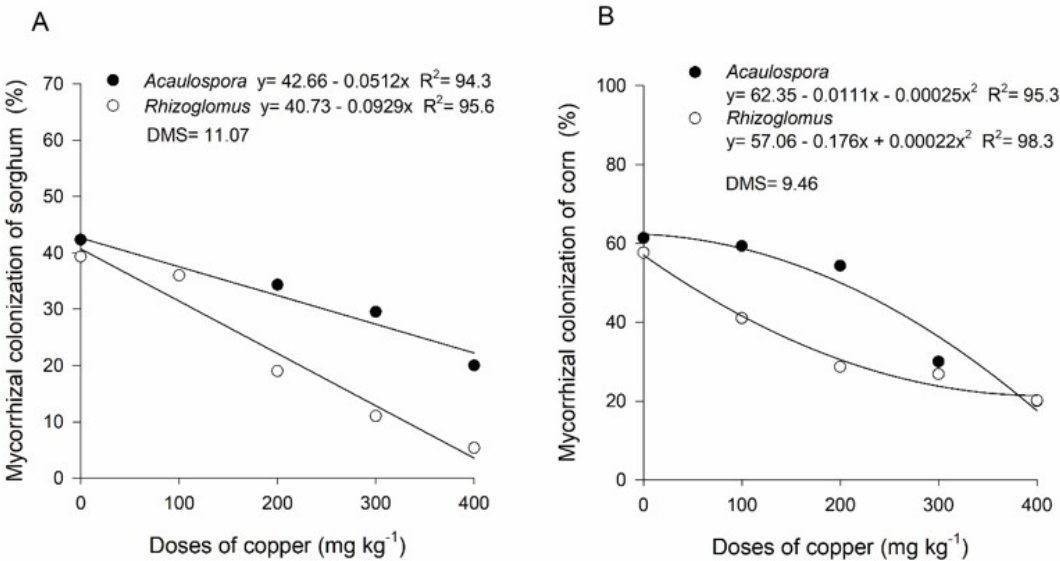


Figure 4. Percentage of mycorrhizal colonization of sorghum and maize grown at different doses of copper, inoculated with *Acaulospora scrobiculata* and *Rhizoglossum clarum*.

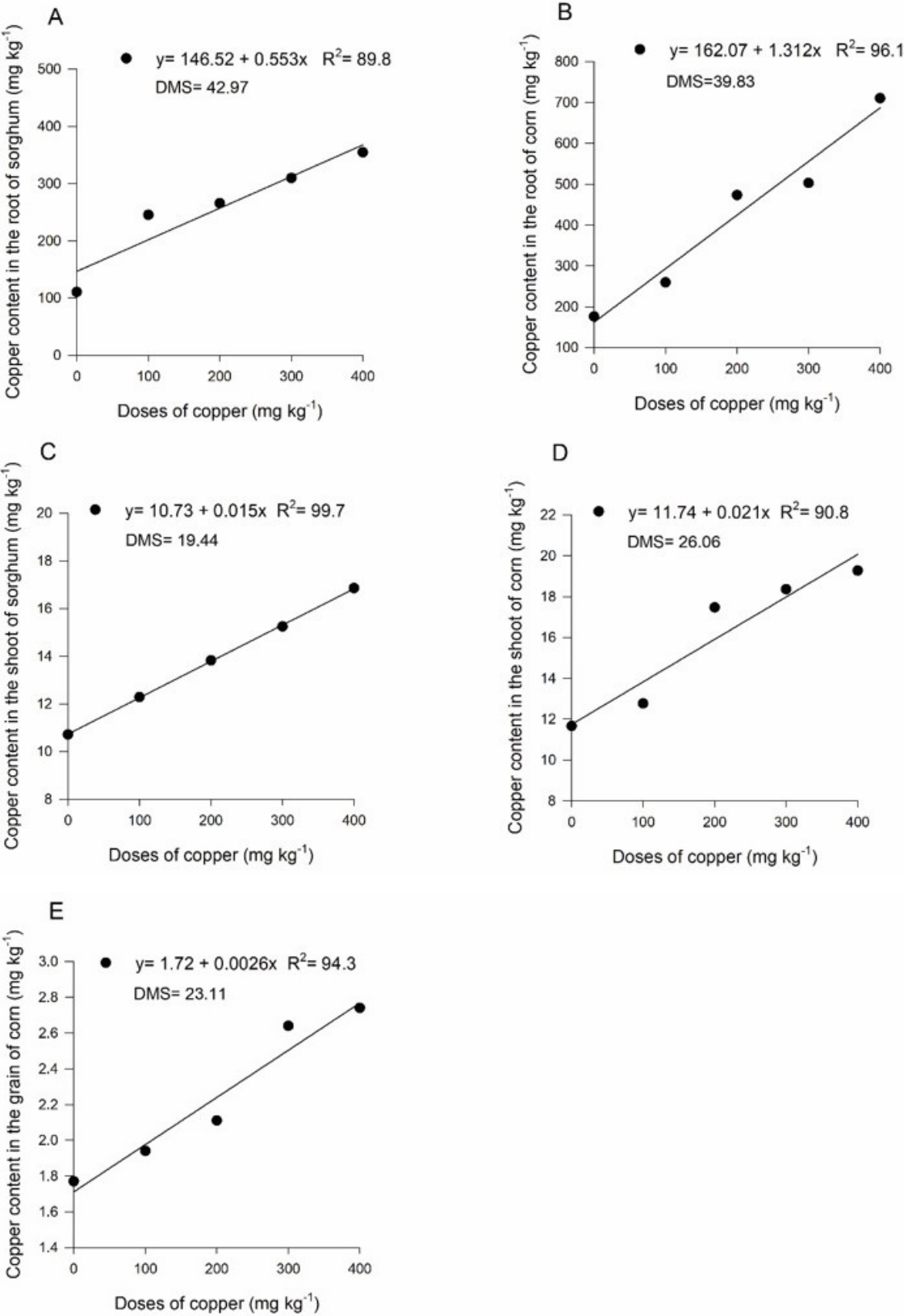


Figure 5. Copper content in the roots (CuR), in the shoot (CuS) of sorghum and corn, and in the grain (CuG) of corn, grown in soil with different doses of copper.

Table 4. Simple effect of inoculation with mycorrhizal fungi on Root copper concentration (RCC) and shoot copper concentration (SCC) of sorghum and corn and grain copper concentration (GCC) of corn.

Inoculum source	-----Sorghum-----		-----Corn-----		
	RCC (mg kg ⁻¹)	SCC (mg kg ⁻¹)	RCC (mg kg ⁻¹)	SCC (mg kg ⁻¹)	GCC (mg kg ⁻¹)
<i>A. scrobiculata</i>	242.71 B*	12.93 B	409.52 B	14.36 B	2.03 A
<i>R. clarum</i>	218.76 B	12.56 B	349.12 B	14.60 B	2.29 A
Control	309.71 A	15.85 A	514.75 A	18.75 A	2.39 A
CV (%)	40.63	18.39	37.67	24.65	21.85

*Means followed by the same capital letter in the column do not differ from each other by the Scott-Knott test ($p < 0.05$) of error probability. CV: coefficient of variation.

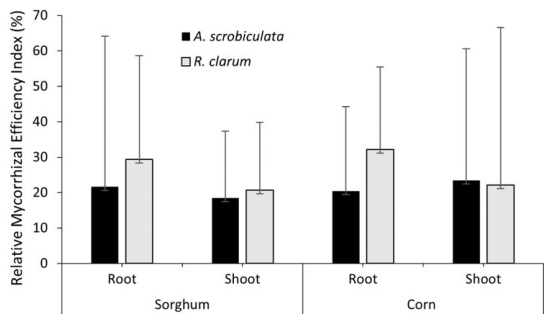


Figure 6. Relative Mycorrhizal Efficiency Index (%) of *Acalospora scrobiculata* and *Rhizoglomus clarum* in reduction copper concentration in shoot and root of *Zea mays* and *Sorghum bicolor*.

doses, presenting 710 mg Cu kg⁻¹ in the root system in the treatment with 400 mg Cu kg⁻¹ (Figure 5A and 5B). In studies carried out by Huerta et al. (2021) and May et al. (2020) it was also found that the highest Cu concentration occurred in the root system of plants, as it was also found in this study.

The shoot Cu concentration of both agricultural species increased linearly with the increase in the applied Cu doses (Figure 5C and 5D), but the levels are below 30 mg Cu kg⁻¹, which is the maximum tolerable limit for food animal (dry matter) (Brasil, 1965). The lower Cu concentration of the shoot compared to the root system was previously reported in other studies (Zancheta et al., 2011; Seidel et al., 2009; Marques et al., 2018). This behavior is related to Cu having affinity for ligands in plant tissues, being retained mainly in the epidermal and cortical cells of the roots (Kopittke et al., 2014), resulting in little mobility within the plant and therefore tends to concentrate in the roots. (Dalcorsio et al., 2014).

The Cu concentration in the corn grain linearly increased with the Cu doses (Figure 5E); however, the levels found were below the maximum tolerance limit proposed by the Brazilian Association of Food Industries (ABIA), i.e. 30 mg Cu kg⁻¹ in grains (ABIA, 1985). Thus, this result did not evidence risks for this product consumption.

Analyzing the simple effect of the mycorrhizal fungi, it was evident that the inoculation significantly reduced the Cu concentration in the roots and shoot of sorghum and corn compared to the control treatment; however,

with no difference between inoculums, while for the Cu concentration in the corn grain there was no difference between treatments (Table 4). Under inoculation, the levels of Cu found are also below the maximum tolerance limit proposed by ABIA (1985) and Brasil (1965).

Analyzing the effect of AMF on Cu phytoremediation in sandy soil, Santana (2018) observed a reduction in Cu levels in the root system of *Canavalia ensiformis* plants when inoculated with *R. clarum*. This is because AMFs are able to accumulate metals in non-vital organs, maintaining the integrity of cell membranes, as these fungi can retain metals in fungal mycelia or cell walls, preventing them from being absorbed by plants (Ruscitti et al., 2017). In addition, due to the greater absorption of phosphorus by AMF (Trovato et al., 2024), there is the formation of phosphate-metal complexes in the roots, allowing a reduction in the mobility and toxicity of this element in the plant (Santana, 2018).

This study evidenced that the mycorrhizal inoculum *A. scrobiculata* and *R. clarum* colonized corn and sorghum roots, being higher in maize and allowing a greater development of these mycorrhized plants compared to the control without inoculation. In addition, inoculation with AMF resulted in lower Cu concentration in shoot and root dry mass of sorghum and corn. AMF, in addition to being fundamental in the absorption of water and nutrients, are also capable of maintaining the growth and development of plants, even in unfavorable environments (Rodrigues et al., 2018).

The relative mycorrhizal efficiency index indicated that both fungi tested were effective in reducing copper concentration in plants, with *R. clarum* showing a slightly higher tendency in root concentration effectiveness in sorghum and corn (Figure 6). Studies confirm that using AMFs can mitigate heavy metal toxicity in plants, an effect linked to antioxidant activation in corn leaves and increased metal retention in roots, in addition to enhancing phosphorus and sulfur uptake (Zhan et al., 2018).

These results demonstrate the importance of using beneficial microorganisms that promote plant growth, such as arbuscular mycorrhizae, which allows satisfactory cultivation and lower health risk in places contaminated with copper, due to the lower concentration of this element in the inoculated plant. However, it is recommended that studies be carried out without sterilizing the soil, avoiding the formation of soluble organic matter that could affect copper availability.

4. Conclusion

The fungal species *A. scrobiculata* allowed greater mycorrhizal colonization in sorghum in soil contaminated with copper.

Inoculation with *A. scrobiculata* and *R. clarum* resulted in greater development and lower Cu concentration in the dry mass of the shoot and root parts of sorghum and corn plants.

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